

International Cybersecurity and Digital Forensics Academy (ICDFA)

CIP-B102: Foundations of Computer and Digital Forensics

LAB 3: Digital Forensics Case Handling, Autopsy and Sleuth Kit Disk Image Analysis

Important Pre-Lab Instruction

Trainees are strongly advised to visit the course materials, read the two Lab 3 slide decks carefully, and understand the concepts before proceeding to this practical lab. Do not start the commands until you understand the investigation steps, chain of custody, forensic image handling, Autopsy workflow, and Sleuth Kit command-line workflow.

Course Code	CIP-B102
Course Title	Foundations of Computer and Digital Forensics
Lab Number	Lab 3
Lab Title	Digital Forensics Image Analysis using Autopsy and The Sleuth Kit
Estimated Duration	4-6 hours practical work plus report writing
Mode	Individual practical lab
Required Evidence Image	Ch01InChap01.dd
Image URL extracted from the slides	https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd

Prepared by: Aminu Idris, AMCPN | Founder, ICDFA

1. Source Materials for Lab 3

- **Slide 5:** Introduction to Digital Forensics - digital forensics definition, NIST view, evidence handling, chain of custody, forensic imaging, Autopsy lab and reporting.
- **Slide 6:** Sleuth Kit Tutorial - Autopsy versus The Sleuth Kit, TSK architecture, image tools, file system tools, deleted file listing, recovery using inode/data units, and mounting a forensic image.

Training DD image URL extracted from the slide decks:

<https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd>

Recommended command-line download URL: <https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd?dl=1>

Reading Requirement Before Lab Execution

Before starting the lab, trainees must read and understand the supplied course materials. The lab is designed to test understanding of digital investigation workflow, not only command memorization. Trainees should be able to explain why each action is performed and how it preserves evidence integrity.

2. Lab Scenario

Scenario: You are acting as a junior digital forensic analyst in the ICDFA lab. You have been provided with a forensic disk image named Ch01InChap01.dd. The requesting officer wants you to identify deleted files, recover relevant office documents, search for useful keywords, preserve your analysis steps, and prepare a repeatable technical report. You must not modify the original evidence image.

3. Learning Outcomes

- Explain the purpose of digital forensics, chain of custody, hashing, and bit-stream image analysis.
- Download and preserve a forensic image in a controlled workspace.
- Calculate and record MD5 and SHA-256 hash values for the evidence image.
- Use Autopsy to create a case, add a data source, locate deleted files, recover files, perform keyword search, tag evidence, and generate a report.
- Use Sleuth Kit command-line tools including `img_stat`, `fsstat`, `fls`, `istat`, `icat`, `blkcat`, `blkls`, and `tsk_recover`.
- Recover INCOME.XLS using at least two command-line approaches and verify file integrity using hashes.
- Prepare a professional forensic lab report with screenshots, commands, observations, and conclusions.

4. Legal, Ethical and Safety Rules

- This lab is for authorized training only. Use only the provided training DD image.
- Do not perform analysis on another person's computer, drive, phone, or files without written authority.
- Keep the original DD image read-only. Conduct recovery and analysis in separate output folders.
- Do not upload real case evidence to online tools. The online hash checking example is only acceptable for this public training image.
- Every screenshot must include the trainee name or terminal path where possible to show originality.

5. Required Tools and Environment

Item	Recommended Tool	Purpose
Operating system	Kali Linux or Ubuntu Linux VM	Run Sleuth Kit commands and manage evidence files
GUI forensic tool	Autopsy	Case creation, deleted file recovery, keyword search and reporting
CLI forensic toolkit	The Sleuth Kit	Low-level image, file system, inode and data-unit analysis
Hashing tools	<code>md5sum</code> , <code>sha256sum</code>	Evidence verification and recovered file comparison
Spreadsheet viewer	LibreOffice Calc or online-safe local viewer	Open recovered .XLS file locally
Internet access	Browser/wget/curl	Download the training DD image

6. Lab Folder Structure

```
mkdir -p ~/CIP-B102-Lab3/{evidence,recovered,reports,screenshots,autopsy-notes}
cd ~/CIP-B102-Lab3
pwd
ls -R
```

Screenshot required: Folder structure showing evidence, recovered, reports, screenshots, and autopsy-notes folders.

7. Evidence Acquisition and Hashing

Download the training forensic image into the evidence folder. The original URL was extracted from the two slide decks.

```
cd ~/CIP-B102-Lab3/evidence
wget -O Ch01InChap01.dd "https://www.dropbox.com/s/nw23q14vzsykyup/Ch01InChap01.dd?dl=1"
ls -lh Ch01InChap01.dd
file Ch01InChap01.dd
```

Download fallback

If wget fails because of browser or Dropbox redirection, open the original URL in a browser, download the file manually, place it inside ~/CIP-B102-Lab3/evidence/, and rename it exactly as Ch01InChap01.dd.

Calculate and save hash values for the original evidence image.

```
md5sum Ch01InChap01.dd | tee ../reports/image_md5.txt
sha256sum Ch01InChap01.dd | tee ../reports/image_sha256.txt
cat ../reports/image_md5.txt
cat ../reports/image_sha256.txt
```

Screenshot required: File size, MD5 hash and SHA-256 hash of Ch01InChap01.dd.

8. Mini Chain-of-Custody Worksheet

Complete this table in your submitted report. This is a training version of evidence documentation.

Field	Student Entry
Case name	
Trainee name	
Date/time evidence was downloaded	
Evidence file name	
Evidence source URL	
Evidence size	
MD5 hash	
SHA-256 hash	
Analysis machine/VM name	
Storage location	

9. Part A - Autopsy GUI Analysis

Use Autopsy to perform a structured graphical investigation of the DD image.

1. Open Autopsy.
2. Create a new case named CIP-B102-Lab3-YourFullName.
3. Use your name as examiner and set the base directory to ~/CIP-B102-Lab3/autopsy-notes or another clear lab folder.
4. Add Ch01InChap01.dd as a data source.
5. Choose the appropriate data source type. If Autopsy asks for image format, choose disk image or raw image as applicable.
6. Run ingest modules suitable for file type identification, hash lookup if available, keyword search, and file analysis.
7. Locate deleted files and record their names, paths, and metadata.
8. Recover at least two deleted files, including INCOME.XLS if available.

9. Search keywords such as income, letter, account, password, and your own chosen keyword.
10. Tag relevant files for reporting and generate an Autopsy report.

Autopsy evidence notes to capture:

- Case name and data source path.
- Evidence image hash shown or independently calculated.
- Deleted file listing with screenshots.
- Recovered file location and file details.
- Keyword search results.
- Generated report screenshot or exported report files.

10. Part B - Sleuth Kit CLI Analysis

Run the following commands from the evidence folder unless otherwise stated. Save important command output in the reports folder.

10.1 Identify the Image

```
cd ~/CIP-B102-Lab3/evidence
img_stat Ch01InChap01.dd | tee ../reports/01_img_stat.txt
mmls Ch01InChap01.dd | tee ../reports/02_mmls.txt
fsstat Ch01InChap01.dd | tee ../reports/03_fsstat.txt
```

If mmls does not show a partition table

Some training images are provided as file system images rather than full disk images with a partition table. If mmls does not show partitions, continue with fsstat, fls, icat and other TSK commands directly against Ch01InChap01.dd.

10.2 List Allocated and Deleted Files

```
fls Ch01InChap01.dd | tee ../reports/04_fls_root.txt
fls -r Ch01InChap01.dd | tee ../reports/05_fls_recursive.txt
fls -r -d Ch01InChap01.dd | tee ../reports/06_deleted_files.txt
cat ../reports/06_deleted_files.txt
```

Answer in your report: How many deleted files are in Ch01InChap01.dd? Show the command output as evidence.

10.3 Find INCOME.XLS and Record Its Metadata Address

```
fls -r Ch01InChap01.dd | grep -i "INCOME"
# Record the metadata address/inode shown beside INCOME.XLS.
# Example format may look like: r/r * 15: INCOME.XLS
```

Replace <INODE> in the commands below with the actual metadata address you found.

10.4 Recover INCOME.XLS Using icat

```
icat Ch01InChap01.dd <INODE> > ../recovered/INCOME_icat.xls
ls -lh ../recovered/INCOME_icat.xls
md5sum ../recovered/INCOME_icat.xls | tee ../reports/income_icat_md5.txt
sha256sum ../recovered/INCOME_icat.xls | tee ../reports/income_icat_sha256.txt
```

10.5 Review File Metadata Using istat

```
istat Ch01InChap01.dd <INODE> | tee ../reports/income_istat.txt
cat ../reports/income_istat.txt
```

From istat, identify the size of the file and the data unit/block numbers used by the file. Include a screenshot of the data unit list.

10.6 Recover INCOME.XLS Using blkcat

Use blkcat to recover the file from its data units. If the file uses one block, use one blkcat command. If it uses multiple blocks, append them in the exact order shown by istat.

```
# One-block example:
blkcat Ch01InChap01.dd <BLOCK_NUMBER> > ../recovered/INCOME_blkcat.xls

# Multi-block example. Replace the block numbers with the actual sequence from istat:
: > ../recovered/INCOME_blkcat.xls
for blk in <BLOCK1> <BLOCK2> <BLOCK3>; do
    blkcat Ch01InChap01.dd "$blk" >> ../recovered/INCOME_blkcat.xls
done

ls -lh ../recovered/INCOME_blkcat.xls
md5sum ../recovered/INCOME_blkcat.xls | tee ../reports/income_blkcat_md5.txt
sha256sum ../recovered/INCOME_blkcat.xls | tee ../reports/income_blkcat_sha256.txt
```

Important

If the blkcat output is larger than the file size, trim only after you understand the file size from istat. Explain any trimming step clearly in your report. Do not silently modify evidence output.

10.7 Recover Files Using tsk_recover

```
mkdir -p ../recovered/tsk_recover_all
tsk_recover -e Ch01InChap01.dd ../recovered/tsk_recover_all
find ../recovered/tsk_recover_all -iname "*INCOME*" -ls | tee ../reports/income_tsk_recover_location.txt
md5sum ../recovered/tsk_recover_all/*INCOME* 2>/dev/null | tee ../reports/income_tsk_recover_md5.txt
```

In your report, compare the hash of the icat-recovered file, blkcat-recovered file, and tsk_recover file. If they are not the same, explain the likely reason.

10.8 View or Convert the Recovered Spreadsheet Locally

```
# Option A: Open in LibreOffice Calc
libreoffice ../recovered/INCOME_icat.xls

# Option B: Convert locally to CSV for safe text viewing
mkdir -p ../recovered/csv
libreoffice --headless --convert-to csv --outdir ../recovered/csv ../recovered/INCOME_icat.xls
ls -lh ../recovered/csv
cat ../recovered/csv/*.csv | tee ../reports/income_csv_preview.txt
```

10.9 Keyword Search and Evidence Notes

```
strings -a Ch01InChap01.dd | grep -i "income" | tee ../reports/search_income.txt
strings -a Ch01InChap01.dd | grep -i "letter" | tee ../reports/search_letter.txt
strings -a Ch01InChap01.dd | grep -i "account" | tee ../reports/search_account.txt
```

Add at least one additional keyword of your own and justify why you selected it.

10.10 Optional Read-Only Mounting

This step is optional. Use it only after you have completed the main Autopsy and TSK recovery steps. Mounting must be read-only.

```
sudo mkdir -p /mnt/cipb102_lab3
sudo mount -o loop,ro ~/CIP-B102-Lab3/evidence/Ch01InChap01.dd /mnt/cipb102_lab3
ls -la /mnt/cipb102_lab3 | tee ~/CIP-B102-Lab3/reports/mounted_listing.txt
sudo umount /mnt/cipb102_lab3
```

11. Required Screenshots Checklist

- ☐ Course materials/pre-lab reading confirmation.
- ☐ Lab folder structure.
- ☐ Downloaded DD image file name, size and location.
- ☐ MD5 and SHA-256 hash of the DD image.
- ☐ Autopsy case creation and data source selection.
- ☐ Autopsy deleted file listing.
- ☐ Autopsy keyword search results.
- ☐ Autopsy tagged evidence and generated report.
- ☐ img_stat, fsstat and fls command outputs.
- ☐ fls -r -d output showing deleted files.
- ☐ INCOME.XLS metadata address identified using fls.
- ☐ icat recovery command and output file.
- ☐ istat details for INCOME.XLS showing size and data units.
- ☐ blkcat recovery attempt and output file.
- ☐ Hash comparison of recovered INCOME.XLS files.
- ☐ Recovered spreadsheet opened or converted to CSV.

12. Student Report Template

11. **1. Cover Page:** Course, lab title, trainee name, date, instructor, and institution.
12. **2. Pre-Lab Confirmation:** State that you read and understood the Introduction to Digital Forensics and Sleuth Kit materials before starting.
13. **3. Case Scenario:** Briefly summarize the scenario in your own words.
14. **4. Evidence Details:** Evidence file name, source URL, file size, MD5 hash, SHA-256 hash, download date/time and storage location.
15. **5. Autopsy Analysis:** Case setup, data source, deleted files found, recovered files, keyword results, tags and report generation.
16. **6. Sleuth Kit Analysis:** Commands used, outputs, deleted file count, INCOME.XLS recovery and metadata analysis.
17. **7. Hash Verification:** Compare recovered files from icat, blkcat and tsf_recover.
18. **8. Findings:** Answer: What was found? Which deleted files were recovered? What was significant?
19. **9. Challenges:** Mention errors, failed commands, redirections, missing tools or corrections made.
20. **10. Conclusion:** Provide a concise forensic conclusion based on evidence, not assumptions.
21. **11. Appendix:** Screenshots and command outputs.

13. Submission Requirements

- One PDF report containing all screenshots and explanations.
- A ZIP file containing reports folder, recovered files folder, and screenshots folder.
- Do not submit the original DD image unless the instructor requests it.
- File naming format: CIP-B102-Lab3-YourFirstName-YourLastName.zip and CIP-B102-Lab3-YourFirstName-YourLastName.pdf.
- Late submission or missing screenshots may reduce marks according to the rubric.

14. Marking Rubric (100 Marks)

Assessment Area	Marks	What the Instructor Will Check
Pre-lab preparation and reading confirmation	5	Student clearly states the materials were read and relates the lab to the concepts.

Assessment Area	Marks	What the Instructor Will Check
Evidence handling and hash documentation	15	Correct folder structure, image source, file size, MD5/SHA-256 and chain-of-custody worksheet.
Autopsy case analysis	20	Case created correctly, image added, deleted files located, keyword search conducted, files tagged/recovered, report generated.
Sleuth Kit command-line analysis	25	Correct use of <code>img_stat</code> , <code>fsstat</code> , <code>fls</code> , <code>istat</code> , <code>icat</code> , <code>blkcat</code> / <code>tsk_recover</code> and saved outputs.
INCOME.XLS recovery and verification	15	Recovered using required methods, hash comparison performed, file content viewed or converted locally.
Screenshots and documentation quality	10	Screenshots are clear, labeled and support the steps claimed.
Final findings and professional conclusion	10	Findings are evidence-based, repeatable and written professionally.

15. Command Reference Summary

Command	Purpose
<code>img_stat image.dd</code>	Show image format information.
<code>mmls image.dd</code>	Display volume/partition layout if present.
<code>fsstat image.dd</code>	Show file system details and statistics.
<code>fls image.dd</code>	List files and directories.
<code>fls -r -d image.dd</code>	Recursively list deleted files.
<code>icat image.dd <inode></code>	Recover file content by metadata address/inode.
<code>istat image.dd <inode></code>	Show metadata details, size and data units for a file.
<code>blkcat image.dd <block></code>	Extract content from a specific data unit/block.
<code>blkls image.dd</code>	Extract or inspect unallocated data units.
<code>tsk_recover -e image.dd output_dir</code>	Recover files, including deleted files, into an output directory.
<code>md5sum / sha256sum</code>	Calculate integrity hashes.
<code>strings image.dd grep keyword</code>	Basic keyword search against printable strings.